other three forces. Over the years, such reasoning has inspired a prodigious and distinguished group of physicists to follow this path vigorously, but the terrain has proven to be fraught with danger, and no one has succeeded in traversing it completely. Let's see why.

General Relativity vs. Quantum Mechanics

The usual realm of applicability of general relativity is that of large, astronomical distance scales. On such distances Einstein's theory implies that the absence of mass means that space is flat, as illustrated in Figure 3.3. In seeking to merge general relativity with quantum mechanics we must now change our focus sharply and examine the microscopic properties of space. We illustrate this in Figure 5.1 by zooming in and sequentially magnifying ever smaller regions of the spatial fabric. At first, as we zoom in, not much happens; as we see in the first three levels of magnification in Figure 5.1, the structure of space retains the same basic form. Reasoning from a purely classical standpoint, we would expect this placid and flat image of space to persist all the way to arbitrarily small length scales. But quantum mechanics changes this conclusion radically. Everything is subject to the quantum fluctuations inherent in the uncertainty principle—even the gravitational field. Although classical reasoning implies that empty space has zero gravitational field, quantum mechanics shows that on average it is zero, but that its actual value undulates up and down due to quantum fluctuations. Moreover, the uncertainty principle tells us that the size of the undulations of the gravitational field gets larger as we focus our attention on smaller regions of space. Quantum mechanics shows that nothing likes to be cornered; narrowing the spatial focus leads to ever larger undulations.

As gravitational fields are reflected by curvature, these quantum fluctuations manifest themselves as increasingly violent distortions of the surrounding space. We see the glimmers of such distortions emerging in the fourth level of magnification in Figure 5.1. By probing to even smaller distance scales, as we do in the fifth level of Figure 5.1, we see that the random quantum mechanical undulations in the gravitational field correspond to such severe warpings of space that it no longer resembles a gently curving geometrical object such as the rubber-membrane analogy used in our discussion in Chapter 3. Rather, it takes on the frothing, turbulent, twisted form illustrated in the